THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** the covered half remains brown with iodine	Light needed.
2	**(c)** parallel to the incident ray but opposite in direction	For 90°, reversed.
3	**(c)** more extreme	Larger variation.
4	**(d)** light energy to chemical energy	Stored in glucose.
5	**(d)** umbra	Complete shadow.
6	**(b)** more frequent leap years	Larger quarter.
7	**(b)** nitrogen-fixing bacteria	Rhizobium.
8	**(b)** virtual image position follows the law of reflection	Geometric consequence.
9	**(a)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
10	**(c)** chlorophyll is needed	Pigment.
11	**(d)** 20°	Twice the turn of the mirror (10 x 2).
12	**(b)** 12 hours	Half.
13	**(b)** chemical energy to usable energy	Released.
14	**(a)** shadow	Dark.
15	**(d)** the Earth moves around the Sun meanwhile	Extra two days to realign.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
17	Angle of incidence is 0° (1); The ray retraces its path (1).	1+1
18	Moon reflects only a small fraction (1); Light is weak (1).	1+1
19	Lack of oxygen (1); They cannot respire (1).	1+1
20	The Sun is low in the evening (1); Rays fall at a smaller angle so the shadow stretches (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
22	i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1).	1+1+1
23	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
24	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
	OR: Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1).	1+1+1
25	Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
	OR: Faster photosynthesis (1); More oxygen (1); Gas escapes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Vary lamp distance (1); Count bubbles per minute (1); Keep temperature and CO2 same (1); Table and graph (1); Rate rises with light (1).	1+1+1+1+1
	OR: Number = 360/60 - 1 = 5 (1); Each mirror forms images (1); Images reflected again (1); Place object and count (1); Vary angle (1).	1+1+1+1+1

27	Fix a stick on the globe (1); Place lamp at the Sun's position (1); Tilt the globe (1); Measure shadows at two latitudes (1); Shadow longer at higher latitudes (1).	1+1+1+1+1
	OR: Set up three conditions (1); Same plant type (1); Starch test after some days (1); Only suitable one forms starch (1); Conclusions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** nitrogen left by pulses	Rhizobium.
2	**(a)** red	Same colour.
3	**(d)** compare dates of nearest distance with seasons	January is nearest.
4	**(b)** NaOH absorbs carbon dioxide	No CO2.
5	**(c)** 0.8 m	Half.
6	**(b)** 182 days	Half.
7	**(c)** more glucose than used in respiration	Growth.
8	**(d)** double	Distance ratio 2:1.
9	**(b)** much nearer to the Earth	It appears equally large in the sky.
10	**(a)** mushrooms have no chlorophyll and absorb food from dead matter	Fungi are saprotrophs.
11	**(c)** black	Absence of light.
12	**(c)** incorrect	Phases are due to viewing angles.
13	**(a)** increases with light intensity	Closer lamp.
14	**(b)** 12 m	2 x 6.
15	**(d)** overhead all year	No tilt.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Earth rotates from west to east (1); Relative motion makes them seem to move opposite (1).	1+1
17	Spines and thick cuticle (1); Fewer stomata (1).	1+1
18	It reflects sunlight (1); Surface scatters light (1).	1+1
19	Sun stays above the horizon longer (1); Tilt effect (1).	1+1
20	No light for photosynthesis (1); Stored starch was used (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Slightly curved (1); Less polished (1); Distorted image (1).	1+1+1
22	Study the Sun's corona (1); Test ideas about light (1); Predict future events (1).	1+1+1
23	Starch increases through the day (1); Photosynthesis exceeds respiration (1); Falls at night (1).	1+1+1
24	Initially 8 m (1); Now 3 m, so 6 m (1); Decrease 2 m (1).	1+1+1
	OR: First quarter about 8 March (1); Full moon about 16 March (1); Last quarter about 23 March (1).	1+1+1
25	Absorb CO2 (1); Store carbon in wood (1); Release oxygen (1).	1+1+1
	OR: Reflect headlights (1); Improve visibility (1); Safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Pinhole real inverted (1); Mirror virtual erect (1); Pinhole needs a screen (1); Mirror image behind (1); Both use straight-line light (1).	1+1+1+1+1
	OR: Regular motions (1); Records of old eclipses (1); Calculations (1); Computer models (1); Accurate predictions (1).	1+1+1+1+1

27	Parasite or saprotroph (1); Check connection to living tissue (1); Observe on dead wood (1); Test for starch (1); Conclude (1).	1+1+1+1+1
	OR: Place object at several distances (1); Find image position by parallax with pin (1); Record distances (1); Compare in a table (1); Equal (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** uses as much oxygen in respiration as it produces in photosynthesis	Balanced gas exchange.
2	**(c)** 50 cm	20 x 100/40.
3	**(a)** about 2.5 times	The Earth's shadow is large.
4	**(c)** saprotroph	Feeds on bread.
5	**(b)** black	Absorbs red.
6	**(c)** incorrect	Far side gets sunlight.
7	**(c)** high humidity	Humid air reduces.
8	**(a)** 2 m behind the mirror	Equal distance.
9	**(b)** every month	Alignment each month.
10	**(a)** photosynthesis is less than respiration	Low light.
11	**(a)** 50 m	40/0.8.
12	**(d)** the Sun is overhead	Equinox.
13	**(c)** Cuscuta, a parasite	Lack chlorophyll.
14	**(c)** red	Reflects red.
15	**(a)** incorrect	It rotates once per orbit.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	5 cm (1); Virtual, erect and laterally inverted (1).	1+1
17	Slight offset from the axis (1); It lies close but not exactly (1).	1+1
18	Dissolves chlorophyll (1); Makes colour change clearer (1).	1+1
19	Image of the Sun on a screen (1); Do not look at the Sun directly (1).	1+1
20	Sun's position changes (1); Earth rotates (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Light travels through more air (1); Blue scatters (1); Red remains (1).	1+1+1
22	Large in shade (1); Small or spines in desert (1); Floating in ponds (1).	1+1+1
23	No light (1); Nothing to reflect (1); Need a source (1).	1+1+1
24	About 12 hours each (1); Little change (1); Sun nearly overhead (1).	1+1+1
	OR: Raise photosynthesis (1); Faster growth (1); Higher yields (1).	1+1+1
25	Height ratio = 15/30 (1); 0.5 x 6 (1); 3 cm (1).	1+1+1
	OR: Summer shadow shorter (1); Sun higher (1); Winter Sun lower (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Narrow shadow (1); Path of totality (1); Safe viewing: pinhole or filters (1); Record time and observations (1); Share with class (1).	1+1+1+1+1
	OR: Photosynthesis takes CO2, gives O2 (1); Respiration reverse (1); Balance in nature (1); Burning adds CO2 (1); Plants reduce it (1).	1+1+1+1+1
27	First mirror shows front (1); Second mirror behind (1); Reflect image of back (1); Repeated images (1); Customer sees the back (1).	1+1+1+1+1

	OR: North of east in summer (1); South of east in winter (1); Due east on equinox (1); Tilt (1); Observation by markers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND